EXERCISE – 3

1. **Write a C program to store n elements in an array using dynamic memory allocation and find largest number.**

```
 #include <stdio.h>
#include <stdlib.h>
int main()
{
   int n, i, largest;
   int *a;
   printf("Enter number of elements: ");
   scanf("%d", &n);
   a = (int *)malloc(n * sizeof(int));
   printf("Enter %d elements:\n", n);
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);
   largest = a[0];
   for(i = 1; i < n; i++)
   {
      if(a[i] > largest)
         largest = a[i];
   }
   printf("Largest element = %d", largest);
   free(a);
   return 0;
}
```

OUTPUT:

```
Enter number of elements: 4
Enter 4 elements:
10 20 30 40
Largest element = 40
```

2. **Print an array in reverse order using pointers**.

```
#include <stdio.h>
int main()
{
   int a[5], i;
   int *p;
  printf("Enter 5 elements:\n");
   for(i = 0; i < 5; i++)
      scanf("%d", &a[i]);
   p = &a[4];
   printf("Array in reverse order:\n");
   for(i = 4; i >= 0; i--)
```

```
    {
        printf("%d ", *p);
        p--;

    return 0;
}
```

OUTPUT:

```
Enter 5 elements:
10 20 30 40 50
Array in reverse order:
50 40 30 20 10
```

3. **Write a C program to store information of students using structures and sort them by name.**

```
#include <stdio.h>
#include <string.h>
struct Student
{
   int rollno;
   char name[50];
   float mark;
};
int main()
{
   struct Student s[3], temp;
   int i, j;
  for(i = 0; i < 3; i++)
   {
      printf("Enter roll no, name and mark: ");
      scanf("%d %s %f", &s[i].rollno, s[i].name, &s[i].mark);
   }
   for(i = 0; i < 2; i++)
   {
      for(j = i + 1; j < 3; j++)
      {
```

```
            if(strcmp(s[i].name, s[j].name) > 0)
            {
               temp = s[i];
               s[i] = s[j];
               s[j] = temp;
            }
         }
      }
      printf("\nStudents sorted by name:\n");
      for(i = 0; i < 3; i++)
      {
         printf("%d %s %.2f\n",
               s[i].rollno, s[i].name, s[i].mark);
      }
      return 0;
}
```

OUTPUT:

```
Enter roll no, name and mark: 10
ANU
490
Enter roll no, name and mark: 94
KENNI
493
Enter roll no, name and mark: 52
CRISS
500

Students sorted by name:
10 ANU 490.00
52 CRISS 500.00
94 KENNI 493.00
```

4. Write a C program to add n distances given in each feet system using structures.

```
#include <stdio.h>
struct Distance
{
   int feet;
};
int main()
{
   struct Distance d[10];
   int n, i, total = 0;
   printf("Enter number of distances: ");
   scanf("%d", &n);
   printf("Enter distances in feet:\n");
   for(i = 0; i < n; i++)
   {
      scanf("%d", &d[i].feet);
      total = total + d[i].feet;
   }
   printf("Total distance = %d feet", total);
   return 0;
}
```

OUTPUT:

```
Enter number of distances: 4
Enter distances in feet:
4
5
7
9
```

Total distance = 25 feet

5. Write a C program to add two complex numbers by passing structure to a function.

```
#include <stdio.h>
struct Complex
{
   int real;
   int imag;
};
struct Complex add(struct Complex c1, struct Complex c2)
{
   struct Complex result;
    result.real = c1.real + c2.real;
   result.imag = c1.imag + c2.imag;
   return result;
}
int main()
{
   struct Complex c1, c2, sum;
   printf("Enter real and imaginary parts of first number: ");
   scanf("%d %d", &c1.real, &c1.imag);
   printf("Enter real and imaginary parts of second number: ");
   scanf("%d %d", &c2.real, &c2.imag);
   sum = add(c1, c2);
   printf("Sum = %d + %di", sum.real, sum.imag);
   return 0;
}
```

OUTPUT:

Enter real and imaginary parts of first number: 6 7

Enter real and imaginary parts of second number: 3 5

Sum = 9 + 12i

6. Write a C program to maintain inventory management system using structure.

```
#include <stdio.h>
struct Product
{
   int id;
   char name[30];
   int quantity;
   float price;
};
int main()
{
   struct Product p[3];
   int i;
   for(i = 0; i < 3; i++)
   {
      printf("Enter product ID, name, quantity and price:\n");
      scanf("%d %s %d %f",
           &p[i].id, p[i].name,
           &p[i].quantity, &p[i].price);
   }
   printf("\nInventory Details:\n");
   for(i = 0; i < 3; i++)
   {
      printf("ID: %d\n", p[i].id);
      printf("Name: %s\n", p[i].name);
      printf("Quantity: %d\n", p[i].quantity);
      printf("Price: %.2f\n\n", p[i].price);
   }
   return 0;
```

```
}
```

OUTPUT:

Enter product ID, name, quantity and price:

101 book 1 10

Enter product ID, name, quantity and price:

102 bag 2 750

Enter product ID, name, quantity and price:

103 pencil 5 7

Inventory Details:

ID: 101

Name: book

Quantity: 1

Price: 10.00

ID: 102

Name: bag

Quantity: 2

Price: 750.00

ID: 103

Name: pencil

Quantity: 5

Price: 7.00

7. **Write a C program for decimal to binary octal and hexadecimal conversion.**

```
#include <stdio.h>
int main()
{
   int n;
   printf("Enter a decimal number: ");
   scanf("%d", &n);
```

```
    printf("Binary = ");
    {
       int a[32], i = 0, temp = n;
       if(temp == 0)
          printf("0");
       while(temp > 0)
       {
          a[i++] = temp % 2;
          temp = temp / 2;
       }

       while(i > 0)
          printf("%d", a[--i]);
    }
    printf("\nOctal = %o", n);
    printf("\nHexadecimal = %X", n);
    return 0;
 }
```

OUTPUT:

```
Enter a decimal number: 35
Binary = 100011
Octal = 43
Hexadecimal = 23
```

8. Write a C program to calculate difference between to time periods.

```
#include <stdio.h>

struct Time

{

   int hours;

   int minutes;

   int seconds;

};

int main()

{

   struct Time t1, t2, diff;

   printf("Enter first time (hours minutes seconds): ");

   scanf("%d %d %d", &t1.hours, &t1.minutes, &t1.seconds);

   printf("Enter second time (hours minutes seconds): ");
```

```
    scanf("%d %d %d", &t2.hours, &t2.minutes, &t2.seconds);
   if(t1.seconds > t2.seconds)
   {
      t2.seconds += 60;
      t2.minutes--;
   }
   diff.seconds = t2.seconds - t1.seconds;
   if(t1.minutes > t2.minutes)
   {
      t2.minutes += 60;
      t2.hours--;
   }
   diff.minutes = t2.minutes - t1.minutes;
   diff.hours = t2.hours - t1.hours;
   printf("Time Difference = %d hours %d minutes %d seconds",
  diff.hours, diff.minutes, diff.seconds);
   return 0;
}
```

OUTPUT:

```
Enter first time (hours minutes seconds): 5 45 50
Enter second time (hours minutes seconds): 8 43 40
Time Difference = 2 hours 57 minutes 50 seconds
```

9. Write a C program to implement an employee management system using structures.

```
#include <stdio.h>
struct Employee
{
   int id;
   char name[30];
   float salary;
};
```

```
int main()
{
   struct Employee e[3];
   int i;
   for(i = 0; i < 3; i++)
   {
      printf("Enter employee ID, name and salary:\n");
      scanf("%d %s %f",
           &e[i].id, e[i].name, &e[i].salary);
   }
   printf("\nEmployee Details:\n");
   for(i = 0; i < 3; i++)
   {
      printf("ID: %d\n", e[i].id);
      printf("Name: %s\n", e[i].name);
      printf("Salary: %.2f\n\n", e[i].salary);
   }
   return 0;
}
```

OUTPUT:

```
Enter employee ID, name and salary:
1 Arun 40000
Enter employee ID, name and salary:
2 bala 50000
Enter employee ID, name and salary:
3 ravi 60000

Employee Details:
ID: 1
Name: Arun
```

Salary: 40000.00

ID: 2

Name: bala

Salary: 50000.00

ID: 3

Name: ravi

Salary: 60000.00

10. Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

```
#include <stdio.h>
void rectangle(float l, float b)
{
   printf("\nRectangle Area = %.2f", l * b);
   printf("\nRectangle Perimeter = %.2f", 2 * (l + b));
}
void triangle(float a, float b, float c)
{
   float s, area;

   s = (a + b + c) / 2;
   area = (s * (s - a) * (s - b) * (s - c));
   printf("\nTriangle Perimeter = %.2f", a + b + c);
   printf("\nTriangle Area = %.2f", area);
}
int main()
{
   float l, b;
   float a, c, d;
```

```
    printf("Enter length and breadth of rectangle: ");
    scanf("%f %f", &l, &b);
    rectangle(l, b);
    printf("\n\nEnter three sides of triangle: ");
    scanf("%f %f %f", &a, &c, &d);
    triangle(a, c, d);
    return 0;
}
```

OUTPUT:

```
Enter length and breadth of rectangle: 8 7

Rectangle Area = 56.00
Rectangle Perimeter = 30.00

Enter three sides of triangle: 9 7 4

Triangle Perimeter = 20.00
Triangle Area = 180.00
```